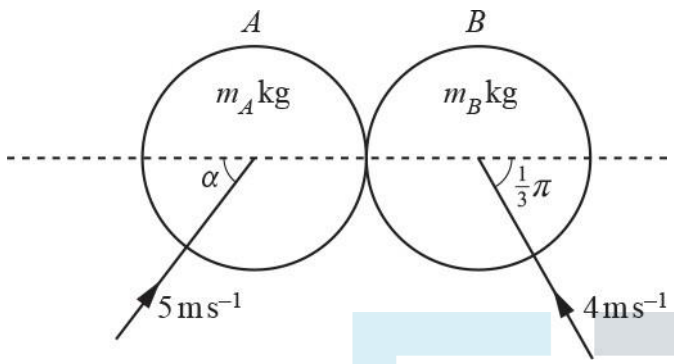
OCR Further Mathematics
Mechanics - Impulse and Momentum
Question Paper
AS and A Level
Further Mathematics A - H235, H245

The latest topic questions for the year 2024 / 2025+ exams.
To be used by all students preparing for OCR AS and A Level Further Mathematics.
Students of other boards may also find this useful.

1(a) Two smooth circular discs A and B of masses m_A kg and m_B kg respectively are moving on a horizontal plane. At the instant before they collide the velocities of A and B are as follows, as shown in the diagram below.

- The velocity of A is 5 ms^{-1} at an angle of α to the line of centres, where $\tan\alpha = \frac{4}{3}$.
- The velocity of B is 4 ms^{-1} at an angle of $\frac{1}{3}\pi$ radians to the line of centres.



The direction of motion of B after the collision is perpendicular to the line of centres.

Show that $\frac{3}{2} \leqslant \frac{m_B}{m_A} \leqslant 4$. [6]

(b) Given that $m_A = 2$ and $m_B = 6$, find the total loss of kinetic energy as a result of the collision. [4]

2(a) A particle P of mass 2 kg is moving on a large smooth horizontal plane when it collides with a fixed smooth vertical wall. Before the collision its velocity is (5 i + 16 j) ms^{-1} and after the collision its velocity is (–3 i + j) ms^{-1}.

The impulse imparted on P by the wall is denoted by I N s.

Find the following.

- The magnitude of I
- The angle between I and i [6]

(b) Find the loss of kinetic energy of P as a result of the collision. [3]

3(a) Two smooth circular discs A and B are moving on a horizontal plane. The masses of A and B are 3 kg and 4 kg respectively. At the instant before they collide

- the velocity of A is $2ms^{-1}$ at an angle of 60° to the line joining their centres,
- the velocity of B is $5ms^{-1}$ towards A along the line joining their centres (see Fig. 6).

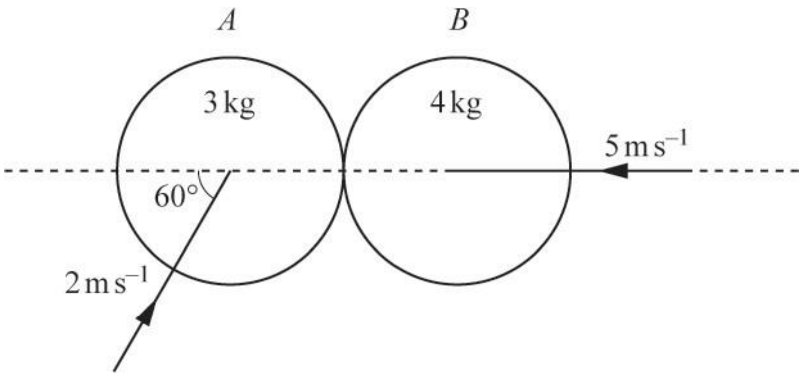


Fig. 6

Given that the velocity of A after the collision is perpendicular to the velocity of A before the collision, find

the coefficient of restitution between A and B, [7]

(b) the total loss of kinetic energy as a result of the collision. [5]

4(a) Two particles A and B, of masses m kg and 1 kg respectively, are connected by a light inextensible string of length d m and placed at rest on a smooth horizontal plane a distance of $\frac{1}{2}$ dm apart. B is then projected horizontally with speed $v$ms^{-1} in a direction perpendicular to AB.

Show that, at the instant that the string becomes taut, the magnitude of the instantaneous impulse in the string, I N s, is given by $I = \frac{\sqrt{3}mv}{2(1+m)}$.

[4]

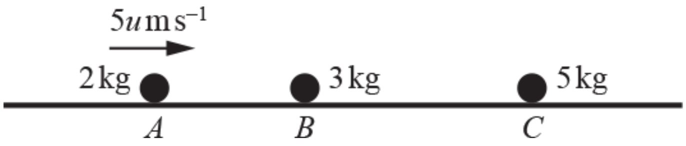
(b) Find, in terms of m and v, the kinetic energy of B at the instant after the string becomes taut. Give your answer as a single algebraic fraction. [3]

(c) In the case where m is very large, describe, with justification, the approximate motion of B after the string becomes taut. [2]

5 Three particles, A, B and C, of masses 2 kg, 3 kg and 5 kg respectively, are at rest in a straight line on a smooth horizontal plane with B between A and C.

Collisions between A and B are perfectly elastic. The coefficient of restitution for collisions between B and C is e.

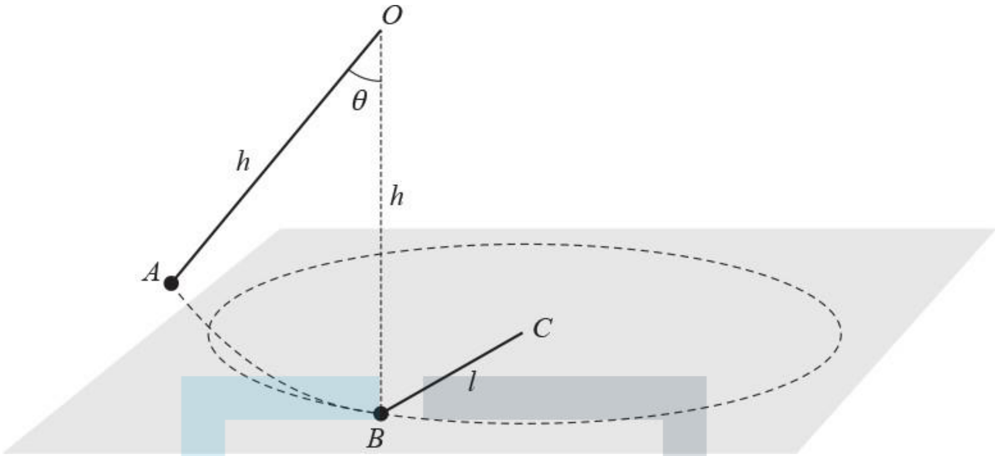
A is projected towards B with a speed of $5u\,\text{ms}^{-1}$ (see diagram).



Show that only two collisions occur.

[8]

6 A point O is situated a distance h above a smooth horizontal plane, and a particle A of mass m is attached to O by a light inextensible string of length h. A particle B of mass $2m$ is at rest on the plane, directly below O, and is attached to a point C on the plane, where $BC = l$, by a light inextensible string of length l. A is released from rest with the string OA taut and making an acute angle θ with the downward vertical (see diagram).



A moves in a vertical plane perpendicular to CB and collides directly with B. As a result of this collision, A is brought to rest and B moves on the plane in a horizontal circle with centre C. After B has made one complete revolution the particles collide again.

(a) Show that, on the next occasion that A comes to rest, the string OA makes an angle ø with the downward vertical through O, where $\cos\phi = \dfrac{3+\cos\theta}{4}$. [9]

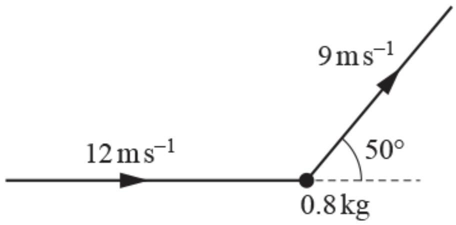
A and *B* collide again when *AO* is next vertical.

(b) Find the percentage of the original energy of the system that remains immediately after this collision. [5]

(c) Explain why the total momentum of the particles immediately before the first collision is the same as the total momentum of the particles immediately after the second collision. [1]

(d) Explain why the total momentum of the particles immediately before the first collision is different from the total momentum of the particles immediately after the third collision. [1]

7 A particle of mass 0.8 kg is moving in a straight line on a smooth horizontal surface with constant speed 12ms^{-1} when it is struck by a horizontal impulse. Immediately after the impulse acts, the particle is moving with speed 9ms^{-1} at an angle of 50° to its original direction of motion (see diagram).



Find

(a) the magnitude of the impulse, [3]

(b) the angle that the impulse makes with the original direction of motion of the particle. [3]

8 A particle P of mass 2.5 kg strikes a rough horizontal plane. Immediately before P strikes the plane it has a speed of $6.5\ \text{m s}^{-1}$ and its direction of motion makes an angle of 30° with the normal to the plane at the point of impact. The impact may be assumed to occur instantaneously. The coefficient of restitution between P and the plane is $\frac{2}{3}$. The friction causes a horizontal impulse of magnitude 2 N s to be applied to P in the plane in which it is moving.

(a) Calculate the velocity of P immediately after the impact with the plane. [7]

(b) P loses about x % of its kinetic energy as a result of the impact. Find the value of x. [2]

9 Two small spheres A and B, of masses 4 kg and 2 kg respectively, are moving in opposite directions along the same straight line towards each other on a smooth horizontal surface. A has speed 1 m s^{-1} and B has speed 3 m s^{-1} before they collide. The coefficient of restitution between A and B is e. The directions of motion of both A and B are reversed as a result of the collision.

(i) Find, in terms of e, the speed of each sphere after the collision and hence show that $e > \frac{1}{8}$. [7]

The total loss in kinetic energy due to the collision is 2.5 J.

(ii) Show that $e = \frac{7}{8}$. [4]

A third small sphere *C* of mass 3 kg is moving in the same straight line as *A* and *B*. After the collision between *A* and *B*, sphere *B* subsequently collides with *C*. The coefficient of restitution between *B* and *C* is *f*, and immediately before this collision *C* is moving with speed 3 m s^{-1} in the opposite direction to *B*.

(iii) The direction of motion of *C* is unchanged by the collision between *B* and *C*, and subsequently *B* collides with *A* again. Find the set of possible values of *f*. [5]

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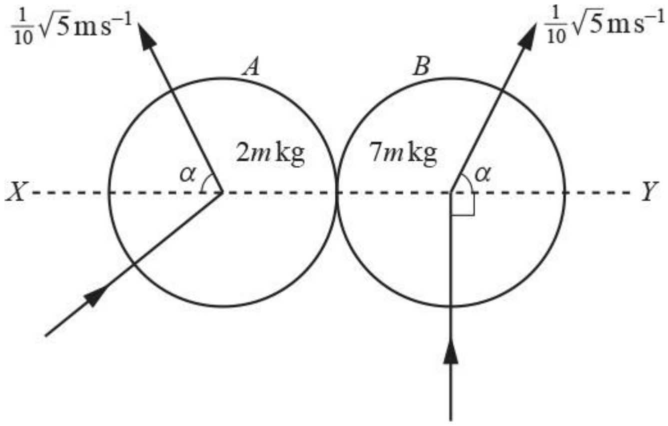

Two uniform smooth spheres, A and B, of equal radius, with masses $2m$ kg and $7m$ kg respectively, collide on a horizontal surface. Immediately before the collision B is moving in a direction perpendicular to the line of centres, XY. Immediately after the collision both A and B are moving with speed $\frac{1}{10}\sqrt{5}\,\mathrm{m\,s^{-1}}$; they are moving on the same side of XY and are both moving in directions making an angle α with XY, where $\tan \alpha = 2$ (see diagram).

(i) Find the speed and direction of motion of A before the collision with B. Find also the coefficient of restitution in the collision. [7]

Subsequently another uniform smooth sphere, C, of mass $3m$ kg and with the same radius as A and B, collides with A. The line of centres of the collision between C and A is parallel to XY; immediately before this collision C is travelling with speed U m s^{-1} from left to right parallel to XY. This collision is perfectly elastic.

(ii) Explain why A will have a second collision with B if U is large enough. [1]

(iii) Find the greatest value of U for which A will not subsequently have a second collision with B. [6]

11 A particle of mass 0.2 kg is moving with speed 4 m s^{-1} in a straight line on a smooth horizontal plane. A horizontal impulse of magnitude 1.2 N s acts on the particle. After the impulse acts the particle is moving with speed 5 m s^{-1}.

Find the angle between the directions of motion of the particle before and after the impulse acts. Find also the angle between the direction in which the impulse acts and the initial direction of motion of the particle. [5]

12 Two uniform smooth spheres *A* and *B* of equal radius are moving on a smooth horizontal surface when they collide. *A* has mass 2.5 kg and *B* has mass 3 kg. Immediately before the collision *A* and *B* each has speed *u* m s^{-1} and each moves in a direction at an angle θ to their line of centres, as indicated in Fig. 1.
Immediately after the collision *A* has speed v_1 m s^{-1} and moves in a direction at an angle α to the line of centres, and *B* has speed v_2 m s^{-1} and moves in a direction at an angle β to the line of centres as indicated in Fig. 2. The coefficient of restitution between *A* and *B* is *e*.

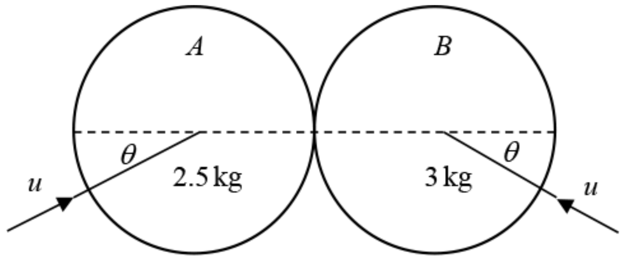

Fig. 1

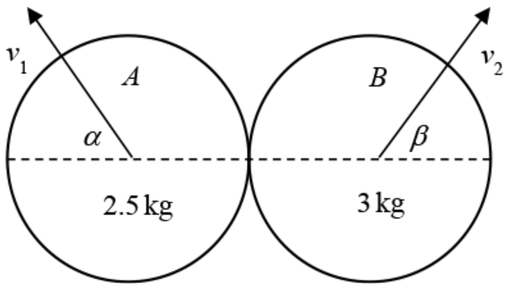

Fig. 2

(a) Show that $\tan \beta = \dfrac{11 \tan \theta}{10e - 1}$. [8]

(b) Given that after impact sphere *A* moves at an angle of 50° to the line of centres and *B* moves perpendicular to the line of centres, find θ. [4]

13 A body, Q, of mass 2 kg moves in a straight line under the action of a single force which acts in the direction of motion of Q. Initially the speed of Q is 5 m s^{-1}. At time t s, the magnitude F N of the force is given by

$$F = t^2 + 3e^t, \ 0 \leq t \leq 4.$$

(a) Calculate the impulse of the force over the time interval. [3]

(b) Hence find the speed of Q when $t = 4$. [2]

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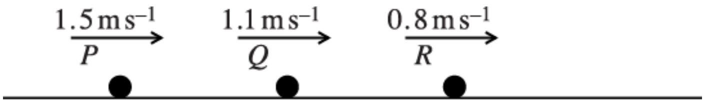

Three particles P, Q and R have masses 0.1 kg, 0.3 kg and 0.6 kg respectively. The particles travel along the same straight line on a smooth horizontal table and have velocities 1.5 m s^{-1}, 1.1 m s^{-1} and 0.8 m s^{-1} respectively (see diagram). P collides with Q and then Q collides with R. In the second collision Q and R coalesce and subsequently move with a velocity of 1 m s^{-1}.

(i) Find the speed of Q immediately before the second collision.

[3]

(ii) Calculate the change in momentum of P in the first collision.

[3]

15 Particle P of mass 0.3 kg and particle Q of mass 0.2 kg are 3.6 m apart on a smooth horizontal surface. P and Q are simultaneously projected directly towards each other along a straight line. Before the particles collide P has speed 4 m s^{-1} and Q has speed 5 m s^{-1}.

(i) Given that the particles coalesce in the collision, calculate their common speed after they collide.

[3]

(ii) It is given instead that one particle is at rest immediately after the collision.

(a) State which particle is in motion after the collision and find the speed of this particle.

[4]

(b) Find the time taken after the collision for the moving particle to return to its initial position.

[4]

(c) On a single diagram sketch the (t, v) graphs for the two particles, with $t = 0$ as the instant of their initial projection.

[4]

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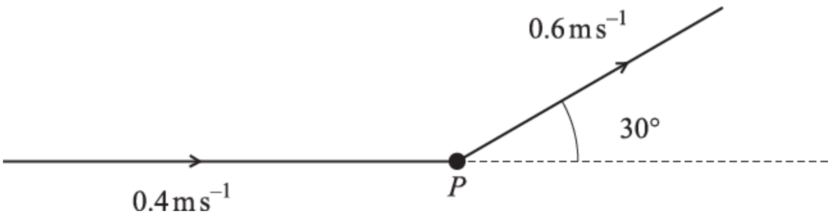

A particle P of mass 0.3 kg is moving with speed 0.4 m s^{-1} in a straight line on a smooth horizontal surface when it is struck by a horizontal impulse. After the impulse acts P has speed 0.6 m s^{-1} and is moving in a direction making an angle 30° with its original direction of motion (see diagram).

(i) Find the magnitude of the impulse and the angle its line of action makes with the original direction of motion of P.

[4]

Subsequently a second impulse acts on P. After this second impulse acts, P again moves from left to right with speed 0.4 m s^{-1} in a direction parallel to its original direction of motion.

(ii) State the magnitude of the second impulse, and show the direction of the second impulse on a diagram.

[2]

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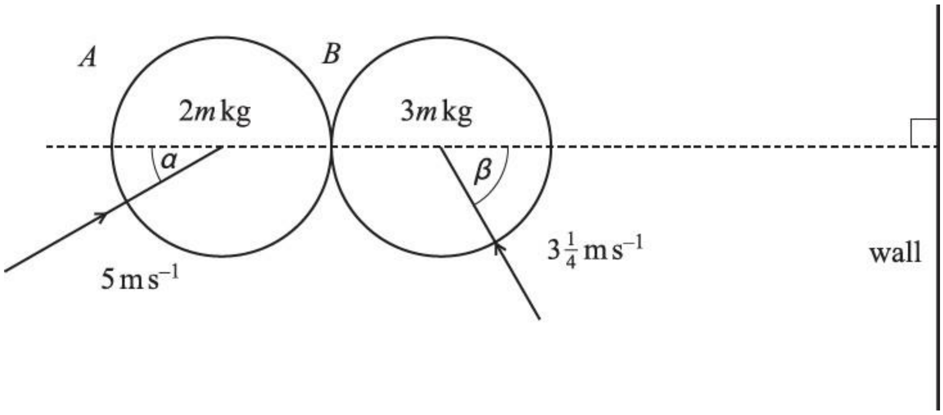


Two uniform smooth spheres *A* and *B*, of equal radius, have masses 2*m* kg and 3*m* kg respectively. The spheres are approaching each other on a horizontal surface when they collide. Before the collision *A* is moving with speed 5 m s^{-1} in a direction making an angle α with the line of centres, where cosθ = $\frac{4}{5}$, and *B* is moving with speed $3\frac{1}{4}$ ms^{-1} in a direction making an angle β with the line of centres, where $\cos\beta = \frac{5}{13}$. A straight vertical wall is situated to the right of *B*, perpendicular to the line of centres (see diagram). The coefficient of restitution between *A* and *B* is $\frac{2}{3}$.

(i) Find the speed of *A* after the collision. Find also the component of the velocity of *B* along the line of centres after the collision.

[7]

B subsequently hits the wall.

(ii) Explain why *A* and *B* will have a second collision if the coefficient of restitution between *B* and the wall is sufficiently large. Find the set of values of the coefficient of restitution for which this second collision will occur.

[3]

18 A particle P of mass 0.2 kg is moving on a smooth horizontal surface with speed 3ms^{-1}, when it is struck by an impulse of magnitude I N s. The impulse acts horizontally in a direction perpendicular to the original direction of motion of P, and causes the direction of motion of P to change by an angle α, where $\tan\alpha = \frac{5}{12}$.

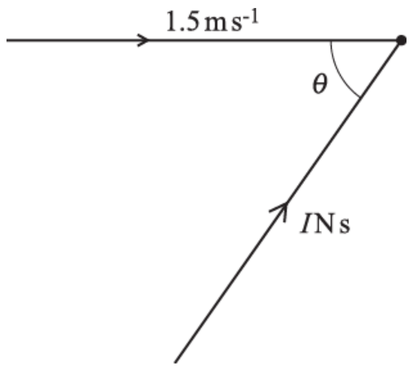
(i) Show that I = 0.25.

[4]

(ii) Find the speed of P after the impulse acts.

[2]

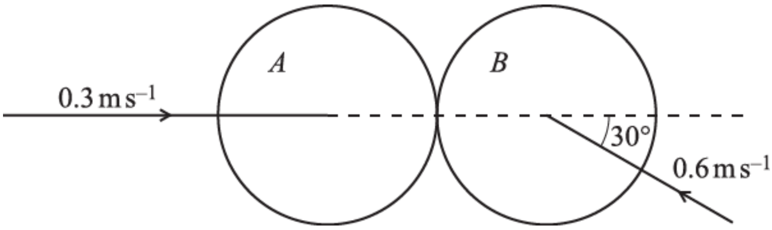
19



A ball of mass 0.6 kg is moving with speed 1.5 m s^{-1} in a straight line. It is struck by an impulse *I* N s acting at an acute angle θ to its direction of motion (see diagram). The impulse causes the direction of motion of the ball to change by an acute angle α, where $\sin\alpha = \frac{8}{17}$. After the impulse acts the ball is moving with a speed of 3.4 m s^{-1}. Find *I* and θ.

[5]

20 Two uniform smooth spheres A and B, of equal radius and equal mass, are moving towards each other on a horizontal surface. Immediately before they collide, A has speed 0.3 m s^{-1} along the line of centres and B has speed 0.6 m s^{-1} at an angle of 30° to the line of centres (see diagram).



After the collision, the direction of motion of B is at right angles to its original direction of motion. Find

(i) the speed of B after the collision,

[3]

(ii) the speed and direction of motion of A after the collision,

[3]

(iii) the coefficient of restitution between A and B.

[3]

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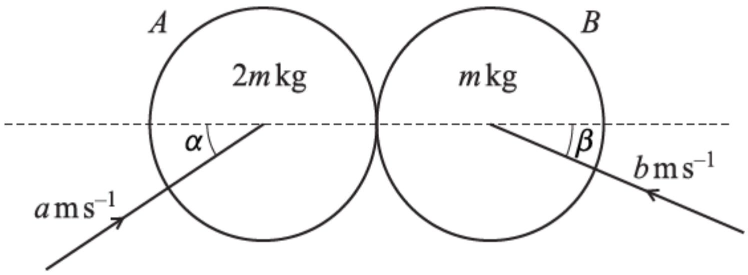


Two uniform smooth spheres A and B, of equal radius, have masses $2m$ kg and m kg respectively. The spheres are moving on a horizontal surface when they collide. Before the collision, A is moving with speed $a$ms^{-1} in a direction making an angle α with the line of centres and B is moving towards A with speed $b$ms^{-1} in a direction making an angle β with the line of centres (see diagram). After the collision, A moves with velocity 2ms^{-1} in a direction perpendicular to the line of centres and B moves with velocity 2ms^{-1} in a direction making an angle of 45° with the line of centres. The coefficient of restitution between A and B is $\frac{2}{3}$.

(i) Show that $a \cos \alpha \frac{5}{6}\sqrt{2}$ and find $b \cos \beta$.

[7]

(ii) Find the values of a and α.

[4]

22 A particle P of mass 0.3 kg is moving on a smooth horizontal surface with speed 0.8 m s^{-1} when it is struck by a horizontal impulse. The magnitude of the impulse is 0.6 N s.

(i)

(a) Find the greatest possible speed of P after the impulse acts.

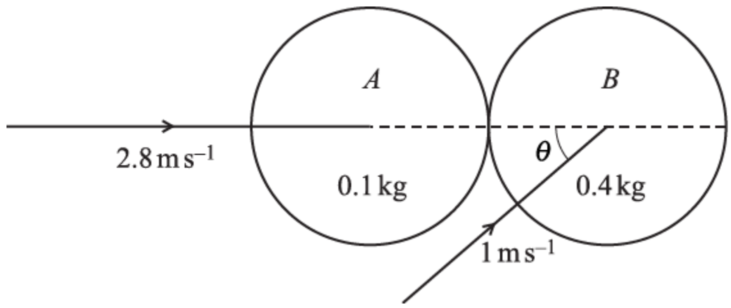
(b) Find the least possible speed of P after the impulse acts.

[3]

(ii) In fact the speed of P after the impulse acts is 2.5 m s^{-1}. Find the angle the impulse makes with the original direction of travel of P and draw a sketch to make this direction clear.

[4]

23



Two uniform smooth spheres A and B of equal radius are moving on a horizontal surface when they collide. A has mass 0.1 kg and B has mass 0.4 kg. Immediately before the collision A is moving with speed 2.8 ms^{-1}along the line of centres, and B is moving with speed 1 m s^{-1} at an angle θ to the line of centres, where cos θ = 0.8 (see diagram). Immediately after the collision A is stationary. Find

(i) the coefficient of restitution between A and B,

[5]

(ii) the angle turned through by the direction of motion of *B* as a result of the collision.

[4]

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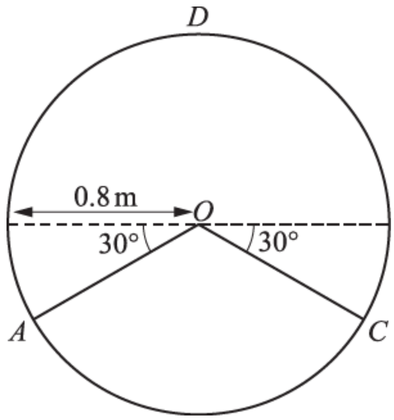


A hollow cylinder is fixed with its axis horizontal. *O* is the centre of a vertical cross-section of the cylinder and *D* is the highest point on the cross-section. *A* and *C* are points on the circumference of the cross-section such that *AO* and *CO* are both inclined at an angle of 30° below the horizontal diameter through *O*. The inner surface of the cylinder is smooth and has radius 0.8 m (see diagram). A particle *P*, of mass *m* kg, and a particle *Q*, of mass 5*m* kg, are simultaneously released from rest from *A* and *C*, respectively, inside the cylinder. *P* and *Q* collide; the coefficient of restitution between them is 0.95.

(i) Show that, immediately after the collision, *P* moves with speed 6.3 m s^{-1}, and find the speed and direction of motion of *Q*.

[8]

(ii) Find, in terms of *m*, an expression for the normal reaction acting on *P* when it subsequently passes through *D*.

[6]

25 A particle of mass 0.3 kg is projected horizontally under gravity with velocity 3.5 m s^{-1} from a point 0.4 m above a smooth horizontal plane. The particle first hits the plane at point *A;* it bounces and hits the plane a second time at point *B*. The distance *AB* is 1 m. Calculate

(i) the vertical component of the velocity of the particle when it arrives at *A*, and the time taken for the particle to travel from *A* to *B*,

[3]

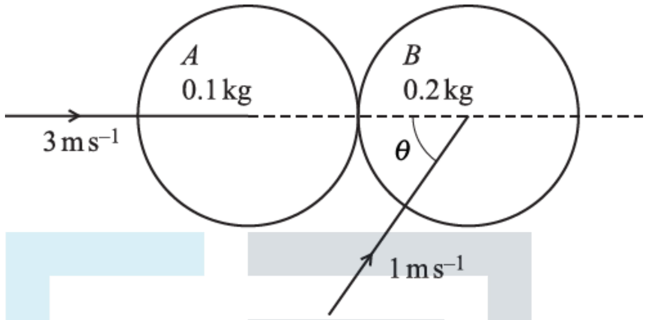
(ii) the coefficient of restitution between the particle and the plane,

[3]

(iii) the impulse exerted by the plane on the particle at A.

[2]

26 Two uniform smooth spheres A and B of equal radius are moving on a horizontal surface when they collide. A has mass 0.1 kg and B has mass 0.2 kg. Immediately before the collision A is moving with speed 3 m s^{-1} along the line of centres, and B is moving away from A with speed 1 m s^{-1} at an acute angle θ to the line of centres, where cosθ = 0.6 (see diagram).



The coefficient of restitution between the spheres is 0.8. Find

(i) the velocity of A immediately after the collision,

[6]

(ii) the angle turned through by the direction of motion of B as a result of the collision.

[5]

27 The masses of two particles A and B are 4 kg and 3 kg respectively. The particles are moving towards each other along a straight line on a smooth horizontal surface. A has speed 8 m s^{-1} and B has speed 10 m s^{-1} before they collide. The kinetic energy lost due to the collision is 121.5 J.

(i) Find the speed and direction of motion of each particle after the collision.

[8]

(ii) Find the coefficient of restitution between A and B.

[2]

28 A small sphere of mass 0.2 kg is projected vertically downwards with a speed of 5 ms^{-1} from a height of 1.6 m above horizontal ground. It hits the ground and rebounds vertically upwards coming to instantaneous rest at a height of h m above the ground. The coefficient of restitution between the sphere and the ground is 0.7.

(i) Find h.

[4]

(ii) Find the magnitude and direction of the impulse exerted on the sphere by the ground.

[3]

(iii) Find the loss of energy of the sphere between the instant of projection and the instant it comes to instantaneous rest at height *h* m.

[3]

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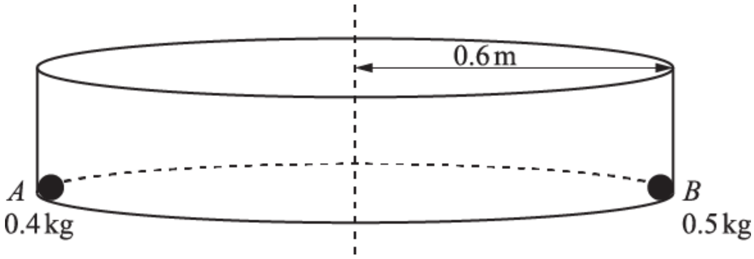

Two small spheres, A and B, are free to move on the inside of a smooth hollow cylinder, in such a way that they remain in contact with both the curved surface of the cylinder and its horizontal base. The mass of A is 0.4 kg, the mass of B is 0.5 kg and the radius of the cylinder is 0.6 m (see diagram). The coefficient of restitution between A and B is 0.35. Initially, A and B are at opposite ends of a diameter of the base of the cylinder with A travelling at a constant speed of v ms^{-1} and B stationary. The magnitude of the force exerted on A by the curved surface of the cylinder is 6 N.

(i) Show that $v = 3$.

[2]

(ii) Calculate the speeds of the particles after *A*'s first impact with *B*.

[6]

Sphere B is removed from the cylinder and sphere A is now set in motion with constant angular speed ω rad s^{-1}. The magnitude of the total force exerted on A by the cylinder is 4.9 N.

(iii) Find ω. [4]

30 Two small spheres A and B, of masses $2m$ kg and $3m$ kg respectively, are moving in opposite directions along the same straight line towards each other on a smooth horizontal surface. A has speed 4 m s^{-1} and B has speed 2 m s^{-1} before they collide. The coefficient of restitution between A and B is 0.4.

(i) Find the speed of each sphere after the collision.

[6]

(ii) Find, in terms of m, the loss of kinetic energy during the collision.

[4]

(iii) Given that the magnitude of the impulse exerted on A by B during the collision is 2.52 N s, find m.

[3]

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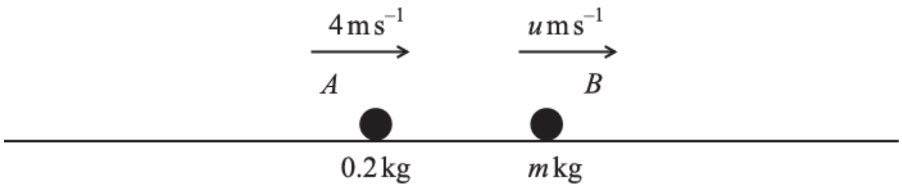


The masses of two particles A and B are 0.2 kg and m kg respectively. The particles are moving with constant speeds 4 ms^{-1} and u ms^{-1} in the same horizontal line and in the same direction (see diagram). The two particles collide and the coefficient of restitution between the particles is e. After the collision, A and B continue in the same direction with speeds $4(1 - e + e^2)$ ms^{-1} and 4 ms^{-1} respectively.

(i) Find u and m in terms of e.

[6]

(ii) Find the value of e for which the speed of A after the collision is least and find, in this case, the total loss in kinetic energy due to the collision.

[5]

(iii) Find the possible values of e for which the magnitude of the impulse that B exerts on A is 0.192 N s.

[2]

32 A particle A is released from rest from the top of a smooth plane, which makes an angle of 30° with the horizontal. The particle A collides 2 s later with a particle B, which is moving up a line of greatest slope of the plane. The coefficient of restitution between the particles is 0.4 and the speed of B immediately before the collision is 2 ms^{-1}. B has velocity 1 ms^{-1} down the plane immediately after the collision. Find

(i) the speed of A immediately after the collision,

[4]

(ii) the distance A moves up the plane after the collision.

[2]

The masses of A and B are 0.5 kg and m kg, respectively.

(iii) Find the value of m. [3]

33 A particle of mass 0.5 kg is held at rest at a point *P*, which is at the bottom of an inclined plane. The particle is given an impulse of 1.8 N s directed up a line of greatest slope of the plane.

(i) Find the speed at which the particle starts to move.

[2]

The particle subsequently moves up the plane to a point Q, which is 0.3 m above the level of P.

(ii) Given that the plane is smooth, find the speed of the particle at Q.

[4]

It is given instead that the plane is rough. The particle is now projected up the plane from *P* with initial speed 3 ms^{-1}, and comes to rest at a point *R* which is 0.2 m above the level of *P*.

(iii) Given that the plane is inclined at 30° to the horizontal, find the magnitude of the frictional force on the particle.

[4]

34

Four particles A, B, C and D are on the same straight line on a smooth horizontal table. A has speed 6 m s^{-1} and is moving towards B. The speed of B is 2 m s^{-1} and B is moving towards A. The particle C is moving with speed 5 m s^{-1} away from B and towards D, which is stationary (see diagram). The first collision is between A and B which have masses 0.8 kg and 0.2 kg respectively.

(i) After the particles collide A has speed 4 m s^{-1} in its original direction of motion. Calculate the speed of B after the collision.

[4]

The second collision is between C and D which have masses 0.3 kg and 0.1 kg respectively.

(ii) The particles coalesce when they collide. Find the speed of the combined particle after this collision.

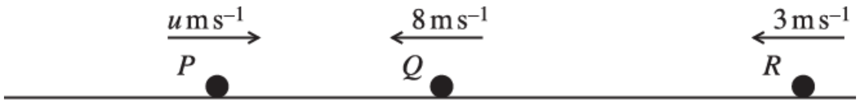
[3]

The third collision is between *B* and the combined particle, after which no further collisions occur.

(iii) Calculate the greatest possible speed of the combined particle after the third collision.

[4]

35



Three particles P, Q and R with masses 0.4 kg, 0.3 kg and m kg are moving along the same straight line on a smooth horizontal surface. P and Q are moving towards each other with speeds u m s^{-1} and 8 m s^{-1} respectively. R has speed 3 ms^{-1} and is moving in the same direction as Q (see diagram).

(i) Immediately after the collision between P and Q their directions of motion have been reversed, but their speeds are unchanged. Calculate u.

[4]

The next collision is between Q and R. After the collision between Q and R, particle Q is at rest and R has speed 9 m s^{-1}.

(ii) Calculate m.

[4]

36

Particles P and Q are moving towards each other with constant speeds 4 ms^{-1} and 2 ms^{-1} along the same straight line on a smooth horizontal surface (see diagram). P has mass 0.2 kg and Q has mass 0.3 kg. The two particles collide.

(i) Show that Q must change its direction of motion in the collision.

[3]

(ii) Given that P and Q move with equal speed after the collision, calculate both possible values for their speed after they collide.

[5]

END OF QUESTION PAPER